Damien MAÎTRE

MSc Engineering Student — CFD, HPC & Numerical Methods

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PROFILE

MSc engineering student specializing in industry, structural mechanics and high performance computing, with hands-on experience across CFD, data analysis and prototype engineering. I'm seeking an engineering internship in the energy or train industry, tackling design, project management or studies, starting March 2027.

EDUCATION

September 2026 – February 2027
MSc Physics engineering
Student exchange at Politecnico di Milano, Italy
International economics, Energy engineering, High performance computing, Infrastructure finance, Neural Networks.

September 2023 – June 2027
MSc Engineering — Fluid dynamics, Structural mechanics, Prototyping
École nationale des ponts et chaussées (ENPC), Paris
Structural mechanics and structure design, FEM, Machine learning, Metallurgy, Complete fluid dynamics module, HPC, Data science.

- **Tools:** OpenFOAM, StarCCM+, Abaqus, OpenSeesPy, gmsh
- **Projects:** Underwater sea-glider simulation and prototype design ; closed-wing aircraft design and optimization ; wooden suspension bridge design, verification and construction.

RESEARCH PROJECTS

Complete CFD developper-first suite
Personal project — OpenFOAM, Gmsh (forked), Python, C++
Open and user-friendly aerodynamical loads studies, tailored for very high $\mathcal{R}e$ axisymmetric problems.

Structural FEM & Eurocode Verification suite
Personal project — OpenSeesPy, Python
End to end structure design, visualization and verification tools.

Data engineering projects
Python, PyTorch, Tensorflow, Numpy, C++ & handmade frameworks
Five years of history working on data analysis and machine learning projects within student teams. City-scale modelisation for mobility and urban optimization ; Genetic algorithm reproducing the swimming behavior of jellyfishes ; Data reporting for large scale engineering studies.

PROFESSIONAL EXPERIENCE

February 2026 – July 2026
Infrastructure engineering internship — TRACTEBEL FRANCE
CDG Express project, Saint-Ouen

- Construction compliance verification for the CDG Express, a 2.5B€ high-speed rail project,
- Structural analysis and Eurocode verifications of civil engineering structures,
- Review of structural calculation reports and as-built site surveys,
- Drafting technical reports and recommendations for the client.

July 2025 – January 2026
Data & Industry Project Manager — SNCF Voyageurs
Industry & digital solutions, Saint-Denis

- Management of a 150k€ digital project within the rolling stock maintenance industry,
- Coordination of 15 3D artists across the country's maintenance plants,
- Project ownership and reporting to internal clients (plant directors, Industrial Director, etc.),
- End-to-end procurement and tendering process.

TECHNICAL SKILLS

CFD & FEM	OpenFOAM, StarCCM+, Abaqus, OpenSeesPy, Gmsh
Programming	10+ years of experience with Python, C++, UNIX environments and various other programming languages.
Prototyping	Additive manufacturing, CAD, Arduino

LANGUAGES

French — Native · **English** — Proficient (C1) · **Italian** — Working proficiency (B1) · **German** — Working proficiency (B2)

ACTIVITIES & COMMUNITY INVOLVEMENT

Rescuer — French Red Cross
Volunteer first-aid responder
First-aid responder at public events (marathons, festivals, Paris-Plage) and on-call with Paris Fire Brigade.

Director — Student Musical Ensemble
BridgeWay Comédie Musicale, Champs-Sur-Marne
Directed an ensemble of 75 artists and a creative/technical leadership team of 12 with 4.5k€ budget.

General Secretary — Student Association
Union des Elèves de l'Ecole Nationale des Ponts et Chaussées
Management of a team of 15 for professional and social events with 1,000+ attendees, Administrative management of a registered French non-profit (Loi 1901).